

The empirical column recurrences for the table OEIS A239178

Adrian Perez Fontelles

Independent researcher

1 September 2026

Abstract

OEIS A239178 is a two-dimensional table $T(n, k)$, and it carries a block of empirical recurrences contributed by R. H. Hardin — one for each of its first few columns. They are true. Column k of the table is an ordinary fixed-width array count, so its rows are the vertices of a finite digraph and $T(n, k)$ is a number of walks in it; being a walk count the column is C -finite, and whether a given recurrence annihilates it is decided by a finite exact computation, with the Cayley–Hamilton theorem bounding the work. This note settles the columns $k = 1$ and the rows $n = 1$.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The table and the conjectures

OEIS A239178 is “ $T(n, k)$ = number of $(n+1) \times (k+1)$ 0..2 arrays with no element greater than all horizontal neighbors or equal to all vertical neighbors.”. It has offset 1, is read by antidiagonals, and begins

6, 26, 12, 98, 82, 36, 378, 514, ...

The entry carries the following block:

Empirical for column k:
k=1: $a(n) = 2 \cdot a(n-1) + 2 \cdot a(n-2)$
k=2: [order 19]
k=3: [order 67]

As of the “Last modified” line on the live entry (Apr 13 2025 08:19:22, revision 15) these are still recorded as empirical, and nothing on the entry records any of them as settled.

2 A column of the table is a walk count

Fix k . The entry defines $T(n, k)$ as the number of arrays of a shape in which one dimension is n and the other is determined by k , subject to a condition imposed on every 2×2 subblock — a condition local to two consecutive rows and two consecutive columns. Substituting the fixed value of k into the entry’s own wording turns the definition into an ordinary one-parameter array count, and for such a count the rows are the vertices of a finite digraph G_k : $r \rightarrow s$ is an edge exactly when placing s under r satisfies the condition in every column pair. An admissible array is then a walk, so with M_k the adjacency matrix of G_k ,

$$T(n, k) = c_k \mathbf{1}^\top M_k^n \mathbf{1}$$

for the constant c_k that the entry’s name supplies (a factor such as $\frac{1}{2}$ or $\frac{1}{4}$ where the name says so, and 1 otherwise).

Lemma 1. Let $q(t) = t^r - \sum_i c_i t^{r-i}$ be the characteristic polynomial of a conjectured recurrence of order r for column k . Then for every $n \geq r$,

$$T(n, k) - \sum_i c_i T(n - i, k) = c_k \mathbf{1}^\top M_k^{n-r} q(M_k) \mathbf{1},$$

so the recurrence holds from some point on if and only if $\mathbf{1}^\top M_k^j q(M_k) \mathbf{1} = 0$ for every $j \geq 0$; and by the Cayley–Hamilton theorem it is enough to check $j < S_k$, where S_k is the number of vertices of G_k .

Proof. Substitute the walk expression and factor M_k^{n-r} out of $M_k^n - \sum_i c_i M_k^{n-i}$; the scalar c_k is nonzero. For the bound, $M_k^{S_k}$ is an integer combination of $I, M_k, \dots, M_k^{S_k-1}$, so each $\mathbf{1}^\top M_k^j q(M_k) \mathbf{1}$ with $j \geq S_k$ repeats that combination of earlier values. $\square$

3 The computation, column by column

line	order	vertices S	proved for	terms checked
column $k=1$	2	12	> 2	8
row $n=1$	6	42	> 6	8

In each case the vector $w = q(M_k) \mathbf{1}$ was formed by r matrix–vector products in exact integer arithmetic and $\mathbf{1}^\top M_k^j w$ evaluated for $j = 0, 1, \dots$, again exactly; every one of those integers vanishes from the stated point onwards and the run continued until the working vector was identically zero, which settles all larger j at once.

Column $k=1$

The sequence is “Number of (n+1) X (2) 0..2 arrays with no element greater than all horizontal neighbors or equal to all vertical neighbors.”. Its rows are the vertices of a digraph on $S = 12$ vertices, edges being the admissible consecutive pairs, and the count is $\mathbf{1}^\top M^n \mathbf{1}$ up to the constant factor in the entry’s name. The conjectured recurrence

$$a(n) = 2a(n-1) + 2a(n-2)$$

holds from index $2 + 1$ on.

Row $n=1$

The sequence is “Number of (2) X (n+1) 0..2 arrays with no element greater than all horizontal neighbors or equal to all vertical neighbors.”. Its rows are the vertices of a digraph on $S = 42$ vertices, edges being the admissible consecutive pairs, and the count is $\mathbf{1}^\top M^n \mathbf{1}$ up to the constant factor in the entry’s name. The conjectured recurrence

$$a(n) = 4a(n-1) - a(n-2) + 3a(n-3) + 3a(n-4) - 4a(n-5) + a(n-6)$$

holds from index $6 + 1$ on.

4 Lines whose recurrence the entry does not print

For the lines below the entry records only an order, the coefficients being in a linked file. They can still be settled, and the recurrences recovered. A line is a walk count on S vertices, so it satisfies some monic recurrence of order at most S and Berlekamp–Massey on $2S$ exact terms returns the MINIMAL one. Where that order equals the stated order — and where the same

- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [3] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009.
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.